MIL-R-11470A(MR)
29 July 1971
SUPERSEDING
MIL-R-11470(Ord)
24 September 1951

MILITARY SPECIFICATION

RADIOGRAPHIC INSPECTION: QUALIFICATION OF EQUIPMENT, OPERATORS AND PROCEDURES

1. SCOPE

1.1 Scope. This specification covers methods for the qualification of equipment, operators and procedures for industrial radiographic inspection.

1.2 Radiographic test kit. An equipment test kit is required to perform the qualification test and may be obtained from the Army Materials and Mechanics Research Center, Watertown, Massachusetts 02172 by submitting a Requisition Invoice Shipping Document, DD Form 1149. Final determination of qualification will be based upon a technical evaluation of test results by the above Center.

1.2.1 Test kit equipment. The test kit supplied will consist of the following:

a. Lead shield
b. Cracked plate "C" (3" x 6" x 1/8")
c. Plate "D" (3" x 6" x 1/8")
d. Plate "E" (3" x 6" x 1/4")
e. Plate "F" (3" x 6" x 1/2")
f. Plate "G" (3" x 6" x 1/2")
g. Plate "H" (3" x 6" x 1/2")
h. Plate "I" (3" x 6" x 1/2")
i. Plate "J" (3" x 6" x 1/4")
j. Plate "K" (5-1/2" x 2-1/4" x 1/4")
k. Two lead supports
l. Quality Assurance Pamphlet AMXMR-P 715-900

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY
MIL-STD-453 - Inspection, Radiographic

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Industrial X-Ray machines with excitation potentials of 440Kv or less.

3.1.1 Proper machine voltage. The X-ray machine shall be checked to ascertain if its voltage is consistant and adequate for the production thickness to be radiographed. The ratings of table I shall be used as standards (see 6.2)

Table I. X-ray machine voltage ratings vs. material thickness

Thickness of steel to be radiographed	Machine rating (kilovolts)
(Inches)	
Not greater than 0.50	100
Not greater than 1.00	150
Not greater than 1.50	200
Not greater than 2.00	250
Not greater than 3.00	300
Not greater than 3.50	400

3.1.2 Working distance (focal distance).

3.1.2.1 Ratio d/t. In order to obtain optimum detail, it is necessary to have a proper working distance for the X-ray tube in relation to the film distance. The working distance (d), and the film distance (t), both measured with reference to the source side of the specimen, are shown in figure 1. This relation of working to film distance (d/t) must be determined for each

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radiographic machine. The method for this determination is given in paragraphs 3.1.2.3 to 3.1.2.4 inclusive.

3.1.2.2 Comparison negative. Using cracked plate C, with the target 30 inches away from the top of the plate, and with the film package in close contact with the underside (see figure 1) the film shall be exposed so as to secure a good radiograph that does not show halation (see 6.3). This radiograph shall be identified as 2A. The above operation should give a picture that shows the best detail obtained with the machine.

3.1.2.3 Determination of optimum value of d/t (see figure 2). With the test kit set up as illustrated in figure 2, a series of radiographs numbered 2B1 through 2B5 shall be made using the values of d and t supplied in table II. These radiographs shall be free from halation (see 6.3).

Table II. X-ray d/t ratios

Identification of radiographs	Relationships (d & t in inches)
2B1	d=15 t=3 d/t=5
2B2	d=30 t=3 d/t=10
2B3	d=40 t=3 d/t=13
2B4	d=50 t=3 d/t=17
2B5	d=60 t=3 d/t=20

3.1.2.4 Selection of optimum d/t ratio. To select this ratio, Radiographs 2B1 through 2B5 shall be compared with Radiograph 2A and the d/t ratio of the radiograph whose image detail is essentially as good as that obtained with comparison Radiograph 2A is to be considered the optimum value of d/t. In cases where it is impossible to employ the determined d/t value for work with the test kit and/or production radiographic work, a higher d/t may be used but under no circumstances shall a value less than the d/t determined by the above operation be used.

3.1.3 Operator qualification.

3.1.3.1 Test to be performed (440 Kv or less. The test kit shall be arranged as indicated in figure 3, with t equal to or greater than the maximum thickness of metal to be radiographed in production work. An appropriate penetrameter (see 3.4) shall be placed on top of cracked plate C in such a position that it does not obscure the crack. Then the proper working distance (d) shall be determined by multiplying the thickness (t) to be radiographed by the optimum working to film distance ratio (d/t) determined in paragraphs 3.1.2.3 and 3.1.2.4; for instance, if t=3 inches and the optimum value of d/t=15, then d=45 inches. A radiograph of the test kit set up in this manner shall be made and identified as 3A.

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3.1.3.2 Image detail. The outline of the penetrameter shall be clearly discernible and the 2-2T hole sharply defined as a round hole when viewed under normal viewing conditions without the aid of magnification. The image detail of cracked plate C shall show with acceptable clarity as indicated by comparison with radiographs of the same test kit which are maintained by the evaluating Center listed in 1.2.1.

3.1.4 Report. A report of the Qualification Test results will be filed as requested in accordance with section 4.

3.2 X-ray machines rated at 1000 Kv.

3.2.1 Proper machine voltage. The practical thickness limits penetrated by 1000 Kv machines for producing acceptable radiographs shall range from a minimum of 1 inch to a maximum of 5 inches.

3.2.2 Working distance (focal distance). The working distance shall be 60 inches or greater when radiographing metal sections of 1 to 5 inches in thickness.

3.2.3 Operator qualification.

3.2.3.1 Test to be performed (1000 Kv). The test kit shall be arranged as indicated in figure 3, with (t) equal to or greater than the maximum thickness of metal to be radiographed in production work and up to but not greater than 5 inches for X-ray machine ratings of 1000 Kv or less. An appropriate penetrameter shall be placed on top of plate (C) in such a position that it does not obscure the crack. Then, using a working distance of 60 inches or more, make a radiograph identified as 3A of the test kit arranged in this manner. Repeat test using the test kit filler plates and cracked plate (C) arranged to equal a total of one inch and identify the radiograph as 3B.

3.2.3.2 Image detail. The image detail shall be as prescribed in paragraph 3.1.3.2.

3.2.4 Report. A report of the Qualification Test results will be filed in accordance with section 4.

3.3 Gamma radiography.

3.3.1 Working distance (focal distance). In order to obtain optimum detail, it is necessary to have a proper working distance for the source of gamma rays in relation to the film distance. The working distance (d) and the film distance (t) are shown in figure 1. This relation of working to film distance (d/t) must be determined for each source of gamma rays. The method for this determination is given in paragraph 3.3.3.

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3.3.2 Comparison negative.

3.3.2.1 Source arrangement. In determining the optimum value of d/t, the gamma ray source shall be arranged with its longest dimension perpendicular to the direction of the radiation that passes through the center of the test kit. Where the source capsule is a cylinder, the diameter of which equals its length, all dimensions shall be regarded as equal and there is no preferred orientation.

3.3.2.2 Exposure. Using plate C, with the source of radiation 30 inches away from the top of the plate, and with the film package in close contact with the underside (see figure 1) the film shall be exposed so as to secure a good radiograph that does not show halation (see 6.3). This radiograph shall be identified as 1-A. The above operation should provide a radiograph that shows the best detail obtainable with the particular source of gamma rays.

3.3.3 Determination of optimum value of d/t (see figure 2). Using values of d and t as shown in table III, films shall be exposed to secure good radiographs free from halation as in the comparison radiograph.

Table III. Gamma-ray d/t ratios

Identification of radiographs	Relationships (d & t in inches)
1B1	d=20 t=5 d/t=4
1B2	d=20 t=2 d/t=10
1B3	d=18 t=1 d/t=18

3.3.4 Selection of optimum d/t ratio. The d/t ratio shall be selected as in 3.1.2.4 except using table III and comparison radiograph 1-A of 3.3.2.2.

3.3.5 Operator qualification.

3.3.5.1 Test to be performed. The test kit shall be arranged as indicated in figure 3, making t=1.25 inches. An appropriate penetrameter shall be placed on top of plate C. Then the proper working distance (d) shall be determined for the thickness (t) to be radiographed by using the optimum value of d/t as determined in paragraph 3.3.4. For example, if t=3 inches and the optimum value of d/t =15, then d=45 inches. A radiograph of the test kit set up in this manner shall be made and identified as 2A.

3.3.5.2 Image detail. Image detail shall be as required by 3.1.3.2.

3.4 Penetrameter design. The penetrameter design to be used throughout the radiographic qualification test procedures outlined herein must be in accordance with MIL-STD-453 or equal.

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3.5 Radiographic density range. All radiographs required in this specification shall be within the H & D density range of 1.50 to 2.50 ± .20.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Machine sources. When penetrating radiation is developed at energies of 1 Mev or less, qualification shall be in accordance with paragraphs 3.1 or 3.2, as applicable with respect to the energy of the radiation involved, such energy being identified by the maximum peak excitation voltage.

4.1.1.1 Isotope sources. Natural or induced radioactive sources that emit gamma rays of energies of 2.19 Mev or less shall be qualified according to paragraph 3.3.

4.2 Method of reporting.

4.2.1 Qualification radiographs and technical data. The Government representative shall be furnished with a complete set of qualification test radiographs together with the applicable, properly completed supplemental data sheets (see pages 11 thru 14) and will forward these to the evaluating Center (see 1.2.1).

5. PREPARATION FOR DELIVERY

Not applicable

6. NOTES

6.1 Intended use. This specification is intended for use with Army contracts involving radiographic requirements.

6.2 Minimum voltages. Table I does not give the correct voltages to be used. It gives the minimum voltages that can be used to radiograph thicknesses as indicated.

6.3 Halation. Fogging of an emulsion due to reflection and dispersion of radiation within the emulsion, usually occurring at locations of heavy exposure, is known as halation.

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6.4 Sources not included. Sources with energies, above 1 Mev for machine type and above 2.19 Mev for isotope type, may be employed and the qualification waived. However, notification giving details for purpose of record is to be furnished to the evaluating Center (see 1.2.1).

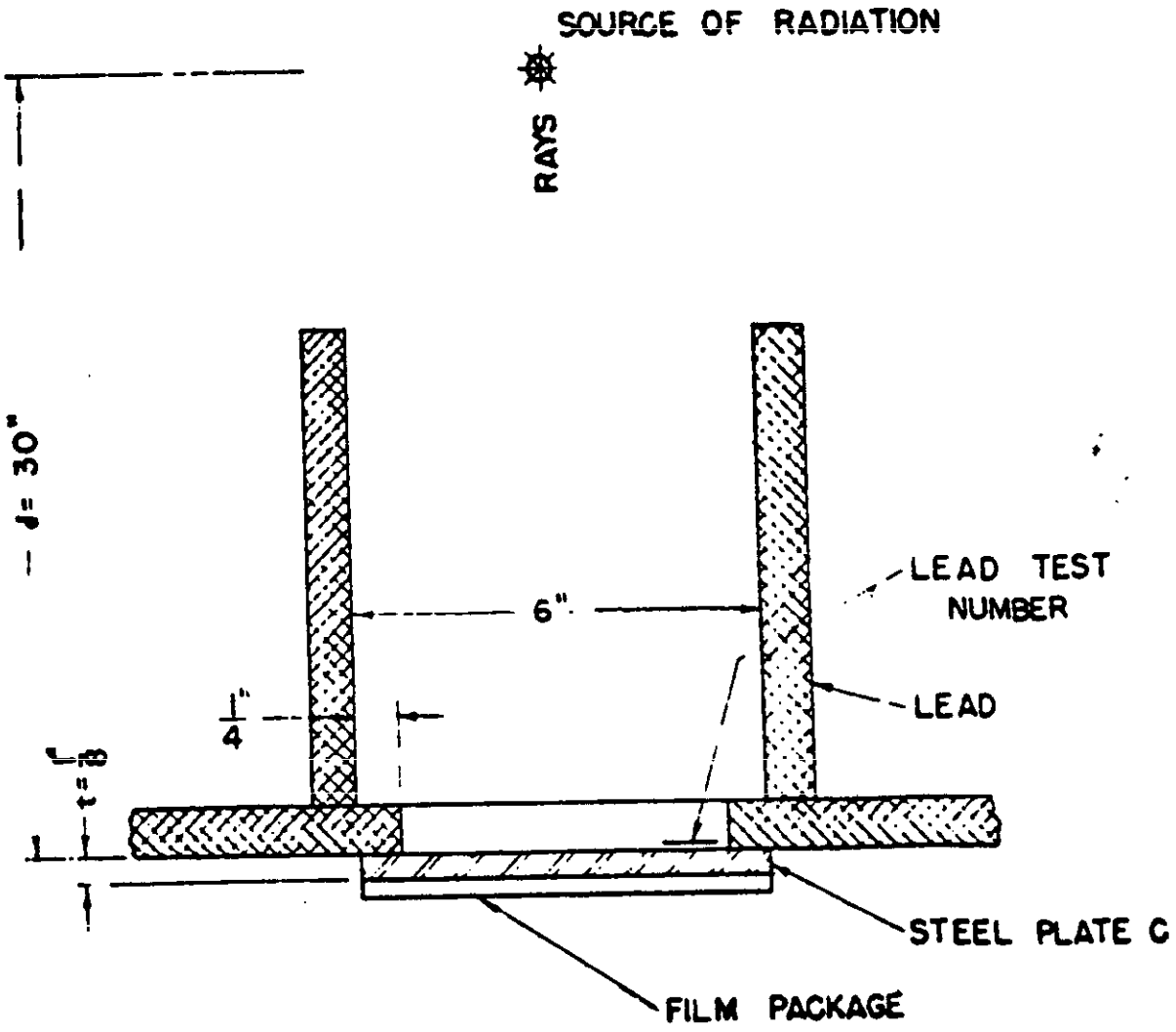
Custodian:
Army-MR

Preparing activity:
Army-MR

User interest:
AT, MI

Project No. MISC-A746

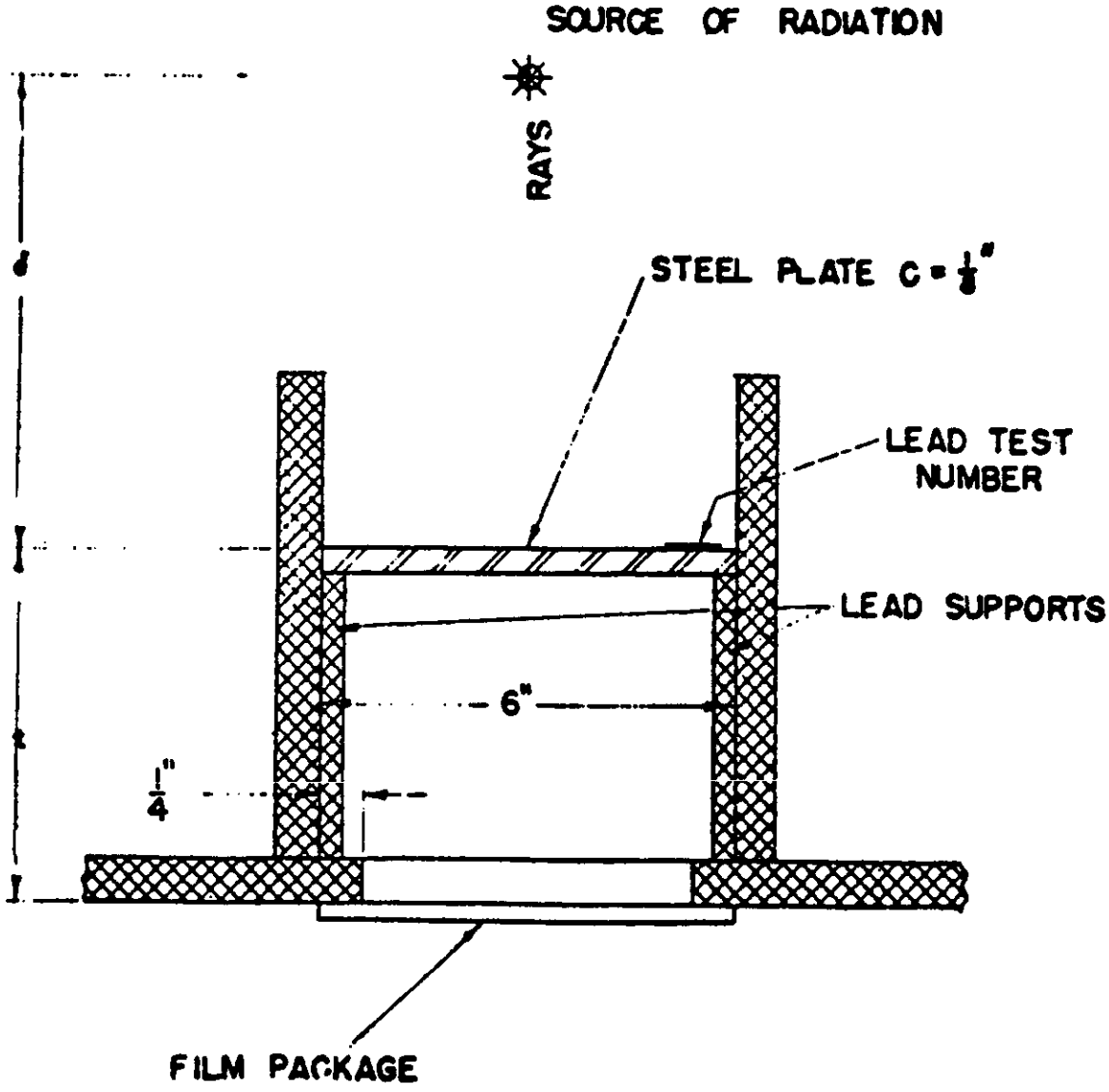
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SECTIONAL VIEW

FIG. 1

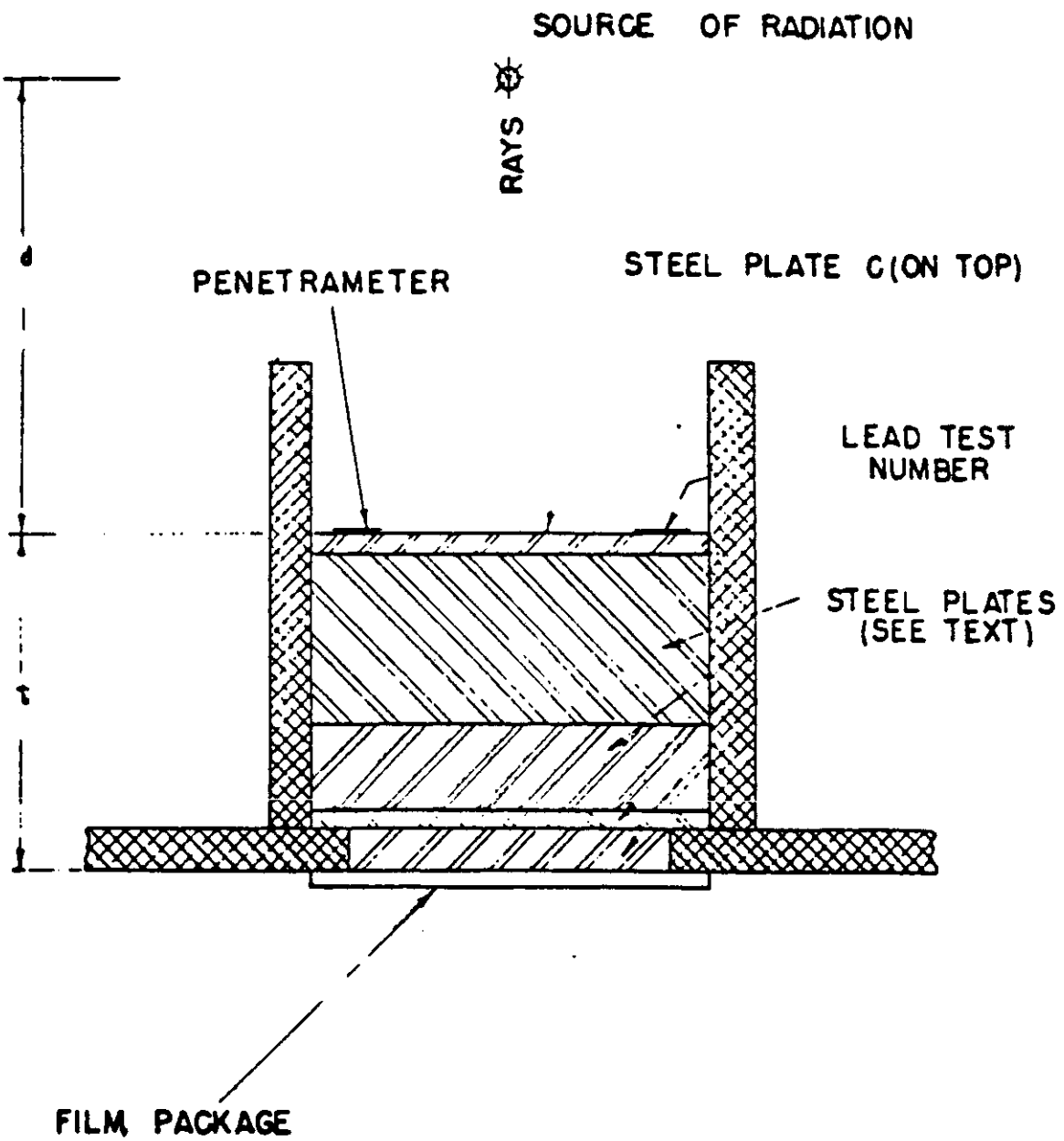
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SECTIONAL VIEW

FIG. 2

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SECTIONAL VIEW

FIG. 3

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SUPPLEMENTAL DATA SHEET

SPECIFICATION MIL-R-11470

X-RAY EQUIPMENT

RADIOGRAPHIC INSPECTION: QUALIFICATION OF EQUIPMENT OPERATORS AND PROCEDURES

FACILITY _____

LOCATION _____ MACHINE MAKE _____

MODEL _____ SERIAL NO. _____ TYPE _____

TUBE TYPE _____ SERIAL NO. _____ TEST BLOCK NO. _____

HAS RADIATION SURVEY BEEN PERFORMED TO DETERMINE FREEDOM FROM HEALTH HAZARDS? YES____ NO____

BY WHOM WAS SURVEY PERFORMED? _____

DATE OF LAST SURVEY _____

SELECTED OPTIMUM D/T RATIO _____ MATERIAL _____ MAXIMUM THICKNESS _____

DATE _____

_____
Government Representatives Signature

_____
Title

_____
Agency or DCAS Office

Remarks

_____

_____

_____

_____

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SUPPLEMENTAL DATA SHEET

OPERATOR _____

TEST DATA

IDENTIFICATION	2A	2B1	2B2	2B3	2B4	2B5	3A	
POTENTIAL (KV.)								
CURRENT (MA.)								
EXPOSURE TIME								
TUBE TO PART DIST. (D)								
PART TO FILM DIST. (T)								
D/T RATIO								
PENETRAMETER								
SCREEN MATERIAL								
SCREEN THICKNESS AT FILM: SOURCE SIDE								
OTHER SIDE								
DEVELOPER (BRAND)								
DEVELOPING TIME								
DEVELOPING TEMP.								
FIXER (BRAND)								
FIXING TIME								
WASHING TIME								
TYPE FILM (BRAND)								
NEGATIVE DENSITY								

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SUPPLEMENTAL DATA SHEET

SPECIFICATION MIL-R-11470

NATURAL AND INDUCED RADIOACTIVE SOURCES

RADIOGRAPHIC INSPECTION: QUALIFICATION OF

EQUIPMENT, OPERATORS AND PROCEDURES

FACILITY_____

LOCATION_____ TYPE OF IOSTOPE_____

A.E.C. LICENSE NO._____ SOURCE OF MANUFACTURE_____

A.E.C. AUTHORIZED PERSON TO HANDLE SOURCE_____

STRENGTH OF SOURCE_____ DATE STRENGTH OF SOURCE DETERMINED_____

PHYSICAL SIZE OF SOURCE: LENGTH_____ DIAMETER_____

SELECTED OPTIMUM D/T RATIO_____ MATERIAL_____ MAXIMUM THICKNESS_____

TEST BLOCK NO._____

DATE_____

_____
Gov't Representatives Signature

_____
Title

_____
Agency or DCAS Office

Remarks

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SUPPLEMENTAL DATA SHEET

OPERATOR _____

TEST DATA							
IDENTIFICATION	2A	2B1	2B2	2B3	2B4	2B5	3A
POTENTIAL (KV.)							
CURRENT (MA.)							
EXPOSURE TIME							
TUBE TO PART DIST. (D)							
PART TO FILM DIST. (T)							
D/T RATIO							
PENETRAMETER							
SCREEN MATERIAL							
SCREEN THICKNESS							
AT FILM: SOURCE SIDE							
OTHER SIDE							
DEVELOPER (BRAND)							
DEVELOPING TIME							
DEVELOPING TEMP.							
FIXER (BRAND)							
FIXING TIME							
WASHING TIME							
TYPE FILM (BRAND)							
NEGATIVE DENSITY							

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DATE

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